

Written Exam Economics summer 2026

Programming for Economists

Date Aug 28 - Aug 30

This exam question consists of 10 pages in total

You should hand-in a single zip-file named with your groupname only. The zip-file should contain:

1. A general README.md for your portfolio
2. Your data analysis project (in the folder 01_dataproject)
3. Your model analysis project (in the folder 02_modelproject)
4. Your exam project (in the folder 03_examproject)

Use of GAI tools is permitted. You must explain how you have used the tools. You are required to complete the GAI declaration and upload it either as a separate attachment or as an integrated part of your exam paper when submitting your assignment. If you do not submit the GAI declaration, your exam submission will be rejected, and you will have used an exam attempt.

When text is solely or mainly generated by a GAI tool, the tool used must be quoted as a source.

Be careful not to cheat at exams!

Exam cheating is for example if you:

- Copy other people's texts without making use of quotation marks and source referencing, so that it may appear to be your own text
- Use the ideas or thoughts of others without making use of source referencing, so it may appear to be your own idea or your thoughts
- Reuse parts of a written paper that you have previously submitted and for which you have received a pass grade without making use of quotation marks or source references (self-plagiarism)
- Receive help from others in violation of the rules in the Faculty of Social Science's common part of the curriculum

You can read more about the rules on exam cheating on your Study Site and in the Faculty of Social Science's common part of the curriculum.

Exam cheating is always sanctioned by a written warning and expulsion from the exam in question. In most cases, the student will also be expelled from the University for one semester.

1 Real GDP across US states

Data for the individual US states is available from **FRED**. For each state, XX , there are two series we need:

1. $XXRGSP$ contains **real GDP** for the state in millions of chained 2017 dollars, at annual frequency.
2. $XXPOP$ contains the **population** of the state in thousands of persons.

The file `states.py` contains the list `STATES` of the 50 two-letter state codes, and a dictionary `REGION` mapping each state to one of the four Census regions (Northeast, Midwest, South and West).

To use FRED you need an API key. A cached copy of the data is provided in the folder `data` in case you cannot get a key in time, but the download itself is part of Question 1.

Throughout, let $y_{i,t}$ be real GDP per person in state i in year t , and let t_{first} and t_{last} be the first and the last year for which you have it.

1.1 Question 1

1. Use an API call to download $XXRGSP$ and $XXPOP$ for all 50 states, and collect each of them in a data frame with the years in the index and the states in the columns.
2. The two series do not cover the same years. Keep only the years where you have both, and drop any state with a missing observation in those years. How many years and how many states are you left with?
3. Compute real GDP per person in dollars, $y_{i,t}$. Be careful with the units: $XXRGSP$ is in millions of dollars and $XXPOP$ is in thousands of persons.
4. Report the highest and the lowest $y_{i,t}$, and which states they belong to, in t_{first} and in t_{last} . Report the ratio between the highest and the lowest in both years, and the average across states in both years.
5. Make a figure with two panels: i) $y_{i,t}$ against t for a handful of states of your own choice, together with the average across states, and ii) $y_{i,t}$ divided by the average across states in the same year, for all states.

1.2 Question 2

1. For each year, compute the standard deviation across states of $\log y_{i,t}$, and plot it against the year.
2. Report its value in t_{first} and in t_{last} , and report in which year it is highest and in which year it is lowest, together with the two values.
3. Describe what the figure shows. Is there a clear trend?

1.3 Question 3

For each state, the average annual growth rate between t_{first} and t_{last} is

$$g_i = \frac{\log y_{i,t_{last}} - \log y_{i,t_{first}}}{t_{last} - t_{first}}$$

From the slope b of a straight line fitted through the points $(\log y_{i,t_{first}}, g_i)$ two further numbers can be computed,

$$\lambda = -\frac{\log(1 + bT)}{T}, \quad h = \frac{\log 2}{\lambda}, \quad T = t_{last} - t_{first}$$

Here λ is the annual speed of convergence. It tells you the proportional rate at which the effect of the initial gap disappears each year. h measures how long it takes for half of an initial gap to disappear.

Note that g_i and b must be measured in log points per year, so if you report g_i in percent, remember to divide the slope by 100 first.

1. Make a scatter plot of g_i against $\log y_{i,t_{first}}$.
2. Fit the straight line $g_i = a + b \log y_{i,t_{first}}$ through the points with `np.polyfit`, and draw it in the same figure. Report a and b , and the correlation between g_i and $\log y_{i,t_{first}}$.
3. Add the names of the five states with the highest g_i and the five states with the lowest g_i to the figure.
4. Report λ and h .
5. Describe the figure. How closely do the points follow the fitted line?

1.4 Question 4

1. Use REGION from `states.py` together with a `groupby` to compute, for each year and each of the four regions, the average across the states in that region of $y_{i,t}$ divided by the average across all states. Report how many states there are in each region.
2. Plot the four series against the year in one figure.
3. Report the value of each of the four series in t_{first} and in t_{last} , and the change between the two years. Which region moves the most, and in which direction?

2 The Solow model with a time-varying savings rate

We consider the **Solow model** from Lecture 9. All variables are per worker. Output is produced with capital,

$$y_t = f(k_t) = k_t^\alpha, \quad 0 < \alpha < 1 \quad (1)$$

A share $s_t \in [0, 1]$ of output is invested and the rest is consumed,

$$i_t = s_t y_t, \quad c_t = (1 - s_t) y_t \quad (2)$$

and capital accumulates as

$$k_{t+1} = i_t + (1 - \delta) k_t \quad (3)$$

Combining the above equations yields the law of motion

$$k_{t+1} = s_t f(k_t) + (1 - \delta) k_t \quad (4)$$

If the savings rate is **constant**, $s_t = s$, then capital settles down at

$$k^* = \left(\frac{s}{\delta}\right)^{\frac{1}{1-\alpha}}, \quad y^* = (k^*)^\alpha, \quad c^* = (1 - s) y^*$$

which is called the *steady state*.

Welfare. The utility of a path for consumption, $c_0, c_1, \dots, c_{T-1}$, is summarized by a single number, the discounted sum of utilities

$$W = \sum_{t=0}^{T-1} \beta^t \log c_t, \quad \beta \in (0, 1) \quad (5)$$

where $\log c_t$ is the per-period utility function. Consumption early counts more than consumption late, and the smaller β is, the more so. A path with a higher W is better than one with a lower W .

The savings rule. Instead of a constant savings rate we let it vary over time as

$$s_t = \bar{s} + (s_0 - \bar{s}) \varphi^t, \quad s_0 \in [0, 1], \quad \varphi \in [0, 1) \quad (6)$$

It starts at s_0 in period 0 and moves back towards $\bar{s}$, at a speed governed by φ . Note that $s_0 = \bar{s}$ gives a constant savings rate whatever φ is.

Calibration. The parameters are

$$\alpha = \frac{1}{3}, \quad \delta = 0.30, \quad \bar{s} = 0.25, \quad \beta = \frac{1}{1.2}, \quad T = 100$$

and the economy always starts with little capital,

$$k_0 = 0.10$$

Code. You can use the code in `SolowModel.py` uploaded together with the exam as your starting point.

2.1 Question 1

1. Compute k^* , y^* and c^* for $s = \bar{s}$ with the formulas above, and confirm k^* with a root-finder for (4) with $k_{t+1} = k_t = k^*$.
2. Simulate the model for T periods with a constant savings rate, $s_t = \bar{s}$, starting from $k_0 = 0.10$. This is the **baseline**. Plot s_t , k_t and c_t against t in three panels. Report c_0 , c_{T-1} and k_{T-1} .
3. Write at least two assert statements testing your simulator on cases where you know the answer in advance. *Hint:* an economy starting in k^* must stay there, and with $s_t = 0$ capital decays as $k_t = (1 - \delta)^t k_0$.

2.2 Question 2

Consider the four combinations

$$(s_0, \varphi) \in \{(0.30, 0.50), (0.40, 0.80), (0.10, 0.50), (0.60, 0.60)\}$$

1. Implement the savings rule s_t in (6).
2. Simulate the model for all four combinations, and plot s_t , k_t and c_t together with the baseline in the same three panels as in Question 1.
3. Report k_{T-1} for the baseline and for all four combinations. Are they the same?
4. Describe in words what s_0 and what φ each do to the paths.

2.3 Question 3

1. Implement W in (5).

2. Report W for the baseline and for each of the four combinations from Question 2, and also the difference $W - W_{\text{baseline}}$, in a small table.
3. Which of the four combinations give a higher W than the baseline? Could you have seen that from the figure in Question 2 alone?

2.4 Question 4

1. Compute W on a grid over $s_0 \in [0, 0.60]$ and $\varphi \in [0, 0.95]$ and illustrate it, for instance with `contourf`. Report the best (s_0, φ) on the grid and the W it gives.
2. Run a numerical optimizer using the best grid point as the initial guess. Report the (s_0, φ) and the W it finds, and compare with the grid.
3. Plot s_t , k_t and c_t under the best rule together with the baseline. Report c_0 and the first period in which c_t is above the baseline.

2.5 Question 5

So far the savings rate has always returned to $\bar{s} = 0.25$ in the long run. Now let the long-run level be a third parameter to choose, so the rule becomes

$$s_t = s_\infty + (s_0 - s_\infty) \varphi^t$$

1. Maximize W over all three parameters (s_0, φ, s_∞) with a numerical optimizer. Report the three parameters and the W they give, and how much W improves on Question 4.
2. Report k_{T-1} for your best three-parameter rule and compare it with k^* from Question 1. Are they the same, and should they be?

2.6 Question 6

The rule above in (6) is only one possible shape of s_t .

1. Propose at least one alternative functional form for s_t of your own. It must settle down at a constant long-run level, so that the economy still ends in a steady state.
2. Implement it and report what W you reach.
3. Put your number next to the ones from Questions 4 and 5 in a small table. Can you beat them?

3 A portfolio with a risky and a safe asset

An investor has wealth W_t , of which a share θ_t is placed in a **risky** asset and $1 - \theta_t$ in a **safe** asset. The gross return on the risky asset is

$$R_t = \exp(\mu + \sigma \varepsilon_t), \quad \varepsilon_t \sim \mathcal{N}(0, 1) \text{ i.i.d.}$$

and the gross return on the safe asset is $R^f = \exp(r)$, the same in every period.

The trading rule. The investor has a *target* share θ^* and a *no-trade band* of width Δ . In each period $t = 0, 1, \dots, T - 1$:

1. If $|\theta_t - \theta^*| > \Delta$ the portfolio is traded all the way back to the target, $\theta_t^{post} = \theta^*$. Otherwise nothing is traded, $\theta_t^{post} = \theta_t$.
2. Trading costs a share τ of the amount traded, so wealth after trading is $W_t^{post} = W_t \left(1 - \tau \left|\theta_t^{post} - \theta_t\right|\right)$.
3. Returns are realized, $W_{t+1} = \theta_t^{post} W_t^{post} R_t + (1 - \theta_t^{post}) W_t^{post} R^f$, and the share in the risky asset at the start of the next period is $\theta_{t+1} = \theta_t^{post} W_t^{post} R_t / W_{t+1}$.

The investor starts at the target, $\theta_0 = \theta^*$, with wealth W_0 .

Note that $\Delta = 0$ means *trade every period* and that $\Delta = 1$ means *never trade*, because θ_t is always between 0 and 1. The same rule therefore covers both extremes.

The goal. The investor only cares about wealth in the final period, and ranks rules by **expected utility**,

$$\mathbb{E}[u(W_T)], \quad u(W) = \frac{W^{1-\gamma}}{1-\gamma}, \quad \gamma > 1$$

which is estimated by the average of $u(W_T)$ across the simulated portfolios. A higher (i.e. less negative) number is better. The best rule is the one that maximizes it.

Calibration. The parameters are

$$\begin{aligned} \mu &= 0.05 & \sigma &= 0.20 & r &= 0.01 \\ \theta^* &= 0.50 & \Delta &= 0.10 & \tau &= 0.01 \\ \gamma &= 3 & W_0 &= 1 & T &= 40 \end{aligned}$$

and $N = 50,000$ is the number of simulated portfolios.

Use the seed 2026 everywhere.

What to report. In Questions 2 and 3 you should report the same **six numbers** for

each rule you look at:

1. the average number of times a portfolio is traded over the T periods
2. the average distance to the target, i.e. the mean of $|\theta_t - \theta^*|$ over both periods and portfolios, measured *before* trading
3. the mean of terminal wealth W_T
4. the median of terminal wealth W_T
5. the 10th percentile of terminal wealth W_T
6. expected utility, $\mathbb{E}[u(W_T)]$

Code. You can use the code in `PortfolioModel.py` uploaded together with the exam as your starting point.

3.1 Question 1

1. Draw ε_t for all periods and all portfolios in one array of shape (N, T) using the seed given, and compute R_t .
2. Report the mean and the standard deviation of $\log R_t$ across all the draws and compare them with μ and σ . Report the mean of R_t and compare it with $\exp\left(\mu + \frac{1}{2}\sigma^2\right)$.
3. Make a figure with two panels: i) a histogram of R_t , and ii) the value of 1 invested in the risky asset in period 0 for 20 of the portfolios, together with the same for the safe asset. Use a logarithmic scale on the y-axis in panel ii).

3.2 Question 2

1. Implement the simulator. Loop over the T periods, but *vectorize* over the N portfolios. Store W_t and θ_t for all periods.
2. Set $\tau = 0$ and simulate the two extreme rules $\Delta = 0$ and $\Delta = 1$. Use the same drawn returns for both. Report the six numbers for each of them.
3. Make a figure with two panels: i) a histogram of W_T for the two rules, and ii) the mean of θ_t against t for the two rules, with a shaded band between the 10th and the 90th percentile.
4. Describe what happens over time to θ_t when the portfolio is never traded.

3.3 Question 3

Set $\tau = 0.01$ and consider the band widths

$$\Delta \in \{0, 0.025, 0.05, 0.075, 0.10, 0.15, 0.20, 0.30, 1\}$$

1. Report the six numbers for each Δ in one table, with Δ in the rows. Use the same drawn returns for all of them.
2. Plot the number of trades and the average distance to the target against Δ , and in a second panel plot expected utility against Δ .
3. Which Δ gives the highest expected utility? By how much does it beat $\Delta = 0$ and $\Delta = 1$?
4. Describe what happens as Δ grows. Which of the six numbers change a lot, and which change very little, when going from $\Delta = 0$ to the best Δ ? Would you have picked the same rule if you had ranked them by the *mean* of W_T instead?